STATIONARY MEASURES OF STOCHASTIC ISING MODELS IN ONE AND TWO DIMENSIONS: AN ALTERNATIVE PROOF OF THE HOLLEY–STROOCK THEOREM

Abstract. Holley and Stroock [HS77] proved that in one and two dimensions every stationary measure of a stochastic Ising model with a finite range potential and strictly positive rates is a Gibbs state. We give a different proof of this result. The law of the spins in a finite volume under a stationary measure is stationary for the finite-volume dynamics with a random boundary condition. Since this dynamics is reversible away from the boundary, its entropy production is bounded by the probability current through flips at the boundary. Instead of a recursion over boxes, we integrate this bound over the level sets of a cut-off function. This shows that, in every dimension, the entropy production of a stationary measure in a finite set $\Lambda \subset \mathbb{Z}^d$ is bounded by a constant times the capacity of Λ in an electrical network on $\mathbb{Z}^d$. In $d \le 2$ these capacities vanish because the network is recurrent. The same argument proves the analogous statement for conservative (exchange) dynamics in $d \le 2$, for stationary measures with finite energy. We also give a non-reversible stationary measure with finite energy for the symmetric exclusion process on a transient subgraph of $\mathbb{Z}^3$. By a coupling with this example, exchange dynamics with summable interactions on the same graph have non-reversible stationary measures as well.

1. Introduction

Let $d \ge 1$ and consider a stochastic Ising model on $E = \{-1,1\}^{\mathbb{Z}^d}$ whose flip rates c_k, $k \in \mathbb{Z}^d$, are associated with a potential J of finite range L in the sense of [HS77]; the assumptions are recalled in Section 2. Neither the potential nor the rates need to be translation invariant. For a probability measure μ on E, let $c_k\mu$ denote the finite measure $c_k(\eta)\,\mu(\mathrm{d}\eta)$, and let $(c_k\mu)\circ\theta_k^{-1}$ denote its image under the map θ_k that flips the spin at k. Then μ is stationary if and only if

$$\sum_{k\in\mathbb{Z}^d} \int f \,\mathrm{d}\big[(c_k\mu)\circ\theta_k^{-1} - c_k\mu\big] = 0 \qquad \text{for all local } f,$$

and μ is a Gibbs state for J if and only if $(c_k\mu)\circ\theta_k^{-1} = c_k\mu$ for every k (Lemma 3.3), that is, if and only if μ is reversible.

Theorem 1.1 (Holley–Stroock [HS77])**.** *Let $d \in \{1,2\}$. A probability measure μ on $\{-1,1\}^{\mathbb{Z}^d}$ is stationary for the stochastic Ising model if and only if it is a Gibbs state for J.*

In other words, in one and two dimensions the vanishing of the sum above forces every summand to vanish. The "if" part is elementary. For the "only if" part, Holley and Stroock use the free energy of $\mu_t = T_t^*\mu$ in a box. For stationary μ its time derivative vanishes, and by a formula of Moulin Ollagnier and Pinchon [MOP77] this derivative equals minus the entropy production in the box plus a term involving only sites near the boundary. Combining the resulting bounds for consecutive boxes leads to a recursive inequality [HS77, (1.22)]. By [HS77, Lemma 1.23], this inequality forces the entropy production to vanish if $\sum_N N^{1-d} = \infty$, i.e. if $d \le 2$.

Our proof does not use the free energy, and it does not use a recursion over boxes. It consists of three steps.

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(i) *Finite-volume dynamics with random boundary conditions.* For finite $\Lambda \subset \mathbb{Z}^d$, the law of η_Λ under a stationary μ is a stationary distribution of a Markov chain on $\{-1,1\}^\Lambda$. In this chain the spins in Λ flip with the rates of the stochastic Ising model, averaged over a random boundary condition drawn from the conditional law of μ given the spins in Λ (Section 3). The measure μ is a Gibbs state if and only if all these chains are reversible. Twice the entropy production of the chain in Λ is

$$\sum_{k\in\Lambda} \Big[H_\Lambda\big(c_k\mu \,\big|\, (c_k\mu)\circ\theta_k^{-1}\big) + H_\Lambda\big((c_k\mu)\circ\theta_k^{-1} \,\big|\, c_k\mu\big)\Big], \tag{1.1}$$

where H_Λ is the relative entropy of the restrictions to the σ-algebra $\mathcal{F}_\Lambda$ generated by the spins in Λ. At sites k whose distance to the complement of Λ is at least L, the rates do not depend on the boundary condition and satisfy detailed balance with respect to the Boltzmann weight in Λ. A general identity for the entropy production of stationary finite Markov chains (Lemma 4.1) then shows that only flips at the remaining sites, which form a set $\partial\Lambda$, contribute. This yields the bound $K\sum_{k\in\partial\Lambda}\|c_k\mu - (c_k\mu)\circ\theta_k^{-1}\|_\Lambda$ for (1.1) (Proposition 4.2). Here $\|\cdot\|_\Lambda$ is the total variation norm on $\mathcal{F}_\Lambda$, and K depends only on the potential and the rates.

(ii) *Integration over level sets.* We apply (i) to all level sets $\{\psi > t\}$, $0 \le t < 1$, of a finitely supported function $\psi : \mathbb{Z}^d \to [0,1]$ with $\psi = 1$ on Λ, and integrate over t with weight $2t\,\mathrm{d}t$. Since relative entropy increases with the σ-algebra, the boundary terms can be absorbed, and (1.1) is at most

$$4K^2 c_+ \sum_{\{k,l\}:\, 0<|k-l|<L} \big(\psi(k)-\psi(l)\big)^2$$

(Theorem 5.1). This holds in every dimension. Taking the infimum over ψ bounds (1.1) by $4K^2c_+\,\mathrm{cap}(\Lambda)$, where $\mathrm{cap}(\Lambda)$ is the capacity of Λ in the electrical network on $\mathbb{Z}^d$ with unit conductances between sites at distance less than L.

(iii) *Recurrence.* If $d \le 2$, this network is recurrent, so $\mathrm{cap}(\Lambda) = 0$ for every finite Λ (Lemma 6.1, where this is checked directly). Hence (1.1) vanishes for every Λ, all finite-volume chains are reversible, and μ is a Gibbs state.

For boxes, the inequality in (i) coincides with the one Holley and Stroock derive from the free energy; Remark 9.1 explains the relation. The main difference is step (ii), which replaces the recursion: the dimension enters only through the recurrence of the network. The proof therefore applies to stochastic Ising models on recurrent graphs (Remark 9.3). For $d \ge 3$ it shows that the entropy production in the box $\{|k| \le n\}$ is $O(n^{d-2})$, the order of the capacity of the box (Remark 9.2).

Sections 7 and 8 treat conservative dynamics, in which neighbouring spins are exchanged. The same argument shows that in $d \le 2$ every stationary measure of an exchange dynamics that has finite energy is reversible (Theorem 7.3). Finite energy takes the place of the positivity of the rates for exchanges across the boundary of a volume. Conversely, the symmetric exclusion process on the subgraph of $\mathbb{Z}^3$ formed by two half-spaces joined by a single bond has a stationary measure that has finite energy but is not reversible (Theorem 8.1). Its density is a non-constant bounded harmonic function, which exists because the graph is transient. A coupling with this example, in the spirit of Lammers and Toninelli [LT24], gives non-reversible stationary measures on the same graph for exchange dynamics whose interactions are summable but otherwise arbitrary (Theorem 8.3).

2. Setting

Configurations. For $k \in \mathbb{Z}^d$ let $|k| := \max_{1\le i\le d} |k_i|$, and let $B_n := \{k \in \mathbb{Z}^d : |k| \le n\}$. Let $E := \{-1,1\}^{\mathbb{Z}^d}$, equipped with the product topology and its Borel σ-algebra $\mathcal{F}$. For $\eta \in E$ and $\Lambda \subseteq \mathbb{Z}^d$, η_Λ is the restriction of η to Λ, and $\mathcal{F}_\Lambda$ is the σ-algebra generated by $\eta \mapsto \eta_\Lambda$. Throughout, Λ denotes a finite subset of $\mathbb{Z}^d$, and $\sum_\sigma$ denotes a sum over all $\sigma \in \{-1,1\}^\Lambda$. For $k \in \mathbb{Z}^d$, η^k is the configuration obtained from η by flipping the spin at k, and $\theta_k : E \to E$ is the map $\eta \mapsto \eta^k$. Similarly, σ^k is obtained from $\sigma \in \{-1,1\}^\Lambda$ by flipping the spin at $k \in \Lambda$. For $k \in \Lambda$ we have $\theta_k^{-1}\{\eta_\Lambda = \sigma\} = \{\eta_\Lambda = \sigma^k\}$. A function on E is *local* if it depends on finitely many coordinates only. For a finite set $F \subset \mathbb{Z}^d$ let $\chi_F(\eta) := \prod_{k\in F} \eta_k$; if $F \subseteq \Lambda$ we also write $\chi_F(\sigma) := \prod_{k\in F} \sigma_k$.

Measures. For a finite measure ν on E and a bounded measurable $f \ge 0$, $f\nu$ is the measure $f(\eta)\,\nu(\mathrm{d}\eta)$, and $\nu \circ \theta_k^{-1}$ is the image of ν under θ_k. For finite measures ν, ν' on E such that $\nu'(\eta_\Lambda = \sigma) > 0$ for all σ we write

$$H_\Lambda(\nu \,|\, \nu') := \sum_\sigma \nu(\eta_\Lambda = \sigma) \log \frac{\nu(\eta_\Lambda = \sigma)}{\nu'(\eta_\Lambda = \sigma)}, \qquad \|\nu - \nu'\|_\Lambda := \sum_\sigma \big|\nu(\eta_\Lambda = \sigma) - \nu'(\eta_\Lambda = \sigma)\big|$$

for the relative entropy of the restrictions of ν and ν' to $\mathcal{F}_\Lambda$ and for the total variation norm of the restriction of $\nu - \nu'$ to $\mathcal{F}_\Lambda$. We use three standard facts. Each holds for finite measures ν, ν' with $\nu(E) = \nu'(E)$, and follows from the case of probability measures by scaling:

(H1) $H_\Lambda(\nu|\nu') \ge 0$, with equality if and only if ν and ν' agree on $\mathcal{F}_\Lambda$;
(H2) $H_\Lambda(\nu|\nu') \le H_{\Lambda'}(\nu|\nu')$ if $\Lambda \subseteq \Lambda'$, by the log-sum inequality;
(H3) $\|\nu - \nu'\|_\Lambda^2 \le 2\,\nu(E)\, H_\Lambda(\nu|\nu')$, by Pinsker's inequality [CT06].

Potential and rates. Fix an integer $L \ge 1$. A *potential with range* L is a family $J = (J_F)$ of real numbers, indexed by the non-empty finite sets $F \subset \mathbb{Z}^d$, such that $J_F = 0$ unless $|k-l| < L$ for all $k, l \in F$, and such that

$$\|J\| := \sup_{k\in\mathbb{Z}^d} \sum_{F\ni k} |J_F| < \infty.$$

Following [HS77, (1.2), (1.3)], rates $(c_k)_{k\in\mathbb{Z}^d}$ are *associated with* J if

$$c_k(\eta) = b_k(\eta) \exp\Big(\sum_{F\ni k} J_F \chi_F(\eta)\Big), \tag{2.1}$$

where each $b_k : E \to (0,\infty)$ depends only on the coordinates η_l with $0 < |l-k| < L$, and $b_- := \inf_k \inf_\eta b_k(\eta) > 0$, $b_+ := \sup_k \sup_\eta b_k(\eta) < \infty$. Then c_k depends only on the coordinates η_l with $|l-k| < L$, and $c_- \le c_k \le c_+$ for $c_\pm := b_\pm e^{\pm\|J\|}$. Since $b_k(\eta^k) = b_k(\eta)$ and $\chi_F(\eta^k) = -\chi_F(\eta)$ for $F \ni k$,

$$\log \frac{c_k(\eta)}{c_k(\eta^k)} = 2\sum_{F\ni k} J_F \chi_F(\eta) \qquad (\eta \in E,\ k \in \mathbb{Z}^d). \tag{2.2}$$

We put

$$K := \log \frac{c_+}{c_-} + 2\|J\|. \tag{2.3}$$

Stationary measures and Gibbs states. The generator acts on local functions f by

$$\mathcal{L}f(\eta) := \sum_{k\in\mathbb{Z}^d} c_k(\eta)\big(f(\eta^k) - f(\eta)\big);$$

the sum has only finitely many non-zero terms.

Definition 2.1. A probability measure μ on $(E,\mathcal{F})$ is *stationary* if $\int \mathcal{L}f \,\mathrm{d}\mu = 0$ for every local function f.

For local f we have $\int c_k\,(f\circ\theta_k)\,\mathrm{d}\mu = \int f\,\mathrm{d}\big((c_k\mu)\circ\theta_k^{-1}\big)$, and therefore

$$\int \mathcal{L}f\,\mathrm{d}\mu = \sum_{k\in\mathbb{Z}^d}\Big[\int f\,\mathrm{d}\big((c_k\mu)\circ\theta_k^{-1}\big) - \int f\,\mathrm{d}(c_k\mu)\Big], \tag{2.4}$$

where only the finitely many k on whose coordinate f depends contribute.

Remark 2.2. Let $(T_t)_{t\ge0}$ be the Feller semigroup of the stochastic Ising model, so that $T_tf - f = \int_0^t T_s\mathcal{L}f\,\mathrm{d}s$ for local f [HS77, (1.5)]. If $T_t^*\mu = \mu$ for all t, then integrating this identity against μ gives $t\int\mathcal{L}f\,\mathrm{d}\mu = 0$. Thus stationarity in the sense of [HS77] implies stationarity in the sense of Definition 2.1. The converse holds as well, because local functions form a core for the generator, but it is not needed here.

Definition 2.3. A probability measure μ is a *Gibbs state for J* if, for every $k\in\mathbb{Z}^d$, the finite measure $\exp\big(\sum_{F\ni k} J_F\chi_F\big)\mu$ is invariant under θ_k.

In [HS77] this invariance is required on the events $\{\eta_\Lambda = \sigma\}$ with $\Lambda \ni k$ finite. Together with $\emptyset$, these events form a π-system that generates $\mathcal{F}$, so the two definitions agree. As noted in [HS77], they are equivalent to the usual definition of Gibbs states via the DLR equations.

3. Finite-volume dynamics with random boundary conditions

For finite Λ let

$$\partial\Lambda := \{k\in\Lambda :\ |l-k| < L \text{ for some } l\in\mathbb{Z}^d\setminus\Lambda\}.$$

If $k\in\Lambda\setminus\partial\Lambda$, then c_k depends only on η_Λ, and we write $c_k(\sigma)$ for its value on $\{\eta_\Lambda=\sigma\}$.

Lemma 3.1 (Balance equations)**.** *Let μ be stationary. Then for every finite Λ and every $\sigma\in\{-1,1\}^\Lambda$,*

$$\sum_{k\in\Lambda}(c_k\mu)(\eta_\Lambda = \sigma^k) = \sum_{k\in\Lambda}(c_k\mu)(\eta_\Lambda=\sigma). \tag{3.1}$$

Proof. Apply (2.4) to $f = \mathbf{1}_{\{\eta_\Lambda=\sigma\}}$. Only $k\in\Lambda$ contribute, and $\big((c_k\mu)\circ\theta_k^{-1}\big)(\eta_\Lambda=\sigma) = (c_k\mu)(\eta_\Lambda=\sigma^k)$. □

Lemma 3.2 (Positivity)**.** *Let μ be stationary. Then $\mu(\eta_\Lambda=\sigma) > 0$ for every finite Λ and every $\sigma\in\{-1,1\}^\Lambda$.*

Proof. Suppose that $\mu(\eta_\Lambda=\sigma)=0$. Then the right-hand side of (3.1) vanishes, so $(c_k\mu)(\eta_\Lambda=\sigma^k)=0$ for every $k\in\Lambda$. Since $c_k\ge c_- > 0$, this gives $\mu(\eta_\Lambda=\sigma^k)=0$ for all $k\in\Lambda$. Iterating, $\mu(\eta_\Lambda=\tau)=0$ for every τ that can be reached from σ by finitely many single flips, i.e. for every $\tau\in\{-1,1\}^\Lambda$. This is impossible. □

Let μ be stationary and Λ finite. By Lemma 3.2, the conditional expectations

$$\mu(c_k \,|\, \eta_\Lambda = \sigma) = \frac{(c_k\mu)(\eta_\Lambda = \sigma)}{\mu(\eta_\Lambda = \sigma)} \in [c_-, c_+] \qquad (k \in \Lambda,\ \sigma \in \{-1,1\}^\Lambda)$$

are well defined. We call the continuous-time Markov chain on $\{-1,1\}^\Lambda$ that jumps from σ to σ^k, $k \in \Lambda$, at rate $\mu(c_k \,|\, \eta_\Lambda = \sigma)$ the *finite-volume dynamics in Λ with random boundary condition*. It describes the spins in Λ flipping with the rates of the stochastic Ising model, averaged over a boundary condition $\eta_{\mathbb{Z}^d\setminus\Lambda}$ drawn from its conditional distribution under μ given $\eta_\Lambda = \sigma$. It has the following properties.

(a) The law $\mu(\eta_\Lambda = \cdot\,)$ of η_Λ under μ is a stationary distribution: (3.1) is the global balance equation

$$\sum_{k\in\Lambda} \mu(\eta_\Lambda = \sigma^k)\, \mu(c_k \,|\, \eta_\Lambda = \sigma^k) = \mu(\eta_\Lambda = \sigma) \sum_{k\in\Lambda} \mu(c_k \,|\, \eta_\Lambda = \sigma).$$

(b) In this stationary state, the probability flux from σ to σ^k is $\mu(\eta_\Lambda = \sigma)\, \mu(c_k \,|\, \eta_\Lambda = \sigma) = (c_k\mu)(\eta_\Lambda = \sigma)$. The flux in the opposite direction is $(c_k\mu)(\eta_\Lambda = \sigma^k) = \big((c_k\mu)\circ\theta_k^{-1}\big)(\eta_\Lambda = \sigma)$. Hence the chain is reversible if and only if $c_k\mu$ and $(c_k\mu)\circ\theta_k^{-1}$ agree on $\mathcal{F}_\Lambda$ for every $k \in \Lambda$.

(c) If $k \in \Lambda \setminus \partial\Lambda$, the rate $\mu(c_k \,|\, \eta_\Lambda = \sigma) = c_k(\sigma)$ does not depend on the boundary condition. By (2.2) it satisfies detailed balance with respect to the free-boundary Boltzmann weight in Λ:

$$c_k(\sigma) \exp\Big(- \sum_{F\subseteq\Lambda} J_F \chi_F(\sigma)\Big) = c_k(\sigma^k) \exp\Big(- \sum_{F\subseteq\Lambda} J_F \chi_F(\sigma^k)\Big), \tag{3.2}$$

because every $F \ni k$ with $J_F \neq 0$ is contained in Λ. Only flips at sites in $\partial\Lambda$ feel the random boundary condition.

Lemma 3.3 (Gibbs states are the reversible measures)**.** *For a stationary measure μ the following are equivalent:*

(i) μ is a Gibbs state for J;
(ii) $(c_k\mu)\circ\theta_k^{-1} = c_k\mu$ for every $k \in \mathbb{Z}^d$;
(iii) for every $n \geq 0$, the finite-volume dynamics in B_n with random boundary condition is reversible.

The equivalence of (i) and (ii) holds for every probability measure μ.

Proof. Fix k. Let $b : E \to (0,\infty)$ be measurable with $b\circ\theta_k = b$. Since θ_k is an involution,

$$\int f \,\mathrm{d}\big((b\nu)\circ\theta_k^{-1}\big) = \int f(\theta_k\eta)\, b(\theta_k\eta)\, \nu(\,\mathrm{d}\eta) = \int f\, b \,\mathrm{d}(\nu\circ\theta_k^{-1})$$

for every finite measure ν and every bounded measurable f, i.e. $(b\nu)\circ\theta_k^{-1} = b\,(\nu\circ\theta_k^{-1})$. Hence, if b and $1/b$ are bounded, $b\nu$ is θ_k-invariant if and only if ν is. With $b = b_k$ and $\nu = \exp(\sum_{F\ni k} J_F\chi_F)\mu$ this shows that (i) and (ii) are equivalent. By (b), (iii) says that $c_k\mu$ and $(c_k\mu)\circ\theta_k^{-1}$ agree on $\mathcal{F}_{B_n}$ for all $n \geq |k|$. For fixed k, the events in $\bigcup_{n\geq|k|}\mathcal{F}_{B_n}$ form a π-system that generates $\mathcal{F}$, and the two measures have the same total mass. So (ii) and (iii) are equivalent. □

Corollary 3.4 (The "if" part of Theorem 1.1)**.** *Every Gibbs state for J is stationary.*

Proof. By Lemma 3.3, every summand on the right-hand side of (2.4) vanishes. □

4. Entropy production of the finite-volume dynamics

We first recall a general fact about stationary Markov chains on a finite state space S. Let $q(x,y) \geq 0$, $x \neq y$, be jump rates such that $q(x,y) > 0$ if and only if $q(y,x) > 0$, and let π be a stationary distribution with $\pi > 0$. The *entropy production* of the chain in the stationary state π is

$$\frac{1}{2} \sum_{x,y:\, q(x,y)>0} \big(\pi(x)q(x,y) - \pi(y)q(y,x)\big) \log \frac{\pi(x)q(x,y)}{\pi(y)q(y,x)}$$

[Sch76]. Each summand is non-negative, and the sum vanishes if and only if the chain is reversible with respect to π.

Lemma 4.1. *In this situation, for every function $V : S \to \mathbb{R}$, the entropy production equals*

$$\frac{1}{2} \sum_{x,y:\, q(x,y)>0} \big(\pi(x)q(x,y) - \pi(y)q(y,x)\big) \Big(\log \frac{q(x,y)}{q(y,x)} - V(x) + V(y) \Big).$$

In particular, the transitions $x \to y$ with $q(x,y)e^{-V(x)} = q(y,x)e^{-V(y)}$, i.e. those satisfying detailed balance with respect to the weight e^{-V}, do not contribute.

Proof. Stationarity of π means $\sum_{y:q(x,y)>0} \big(\pi(x)q(x,y) - \pi(y)q(y,x)\big) = 0$ for every x. Multiplying by $f(x)$, summing over x and symmetrising gives

$$\frac{1}{2} \sum_{x,y:\, q(x,y)>0} \big(\pi(x)q(x,y) - \pi(y)q(y,x)\big)\big(f(x) - f(y)\big) = 0$$

for every $f : S \to \mathbb{R}$. Take $f = \log \pi + V$ and subtract this from the entropy production, using $\log \frac{\pi(x)q(x,y)}{\pi(y)q(y,x)} = \log \frac{\pi(x)}{\pi(y)} + \log \frac{q(x,y)}{q(y,x)}$. □

We apply this to the finite-volume dynamics in Λ with random boundary condition, in its stationary state $\mu(\eta_\Lambda = \cdot\,)$. By property (b) of Section 3, its entropy production is

$$\begin{aligned} &\frac{1}{2} \sum_{k\in\Lambda} \sum_{\sigma} \Big[(c_k\mu)(\eta_\Lambda = \sigma) - (c_k\mu)(\eta_\Lambda = \sigma^k)\Big] \log \frac{(c_k\mu)(\eta_\Lambda = \sigma)}{(c_k\mu)(\eta_\Lambda = \sigma^k)} \\ &\qquad = \frac{1}{2} \sum_{k\in\Lambda} \Big[H_\Lambda\big(c_k\mu \,\big|\, (c_k\mu)\circ\theta_k^{-1}\big) + H_\Lambda\big((c_k\mu)\circ\theta_k^{-1} \,\big|\, c_k\mu\big)\Big], \end{aligned} \tag{4.1}$$

and each summand on the right is non-negative by (H1). By Lemma 3.3 and (H1), a stationary μ is a Gibbs state if and only if

$$\sum_{k\in B_n} \Big[H_{B_n}\big(c_k\mu \,\big|\, (c_k\mu)\circ\theta_k^{-1}\big) + H_{B_n}\big((c_k\mu)\circ\theta_k^{-1} \,\big|\, c_k\mu\big)\Big] = 0 \qquad \text{for all } n \geq 0. \tag{4.2}$$

Proposition 4.2 (Boundary estimate)**.** *Let μ be stationary and let $\Lambda \subset \mathbb{Z}^d$ be finite and non-empty. Then*

$$\sum_{k\in\Lambda} \Big[H_\Lambda\big(c_k\mu \,\big|\, (c_k\mu)\circ\theta_k^{-1}\big) + H_\Lambda\big((c_k\mu)\circ\theta_k^{-1} \,\big|\, c_k\mu\big)\Big] \leq K \sum_{k\in\partial\Lambda} \big\| c_k\mu - (c_k\mu)\circ\theta_k^{-1} \big\|_\Lambda.$$

Proof. Apply Lemma 4.1 to the finite-volume dynamics in Λ with $V(\sigma) := \sum_{F\subseteq\Lambda} J_F\chi_F(\sigma)$. By (3.2), the transitions $\sigma \to \sigma^k$ with $k \in \Lambda \setminus \partial\Lambda$ do not contribute. For $k \in \partial\Lambda$, the factor in brackets is

$$r_k(\sigma) := \log \frac{\mu(c_k \,|\, \eta_\Lambda = \sigma)}{\mu(c_k \,|\, \eta_\Lambda = \sigma^k)} - 2 \sum_{F\subseteq\Lambda,\, F\ni k} J_F\chi_F(\sigma),$$

and $|r_k(\sigma)| \le \log(c_+/c_-) + 2\|J\| = K$. By property (b) of Section 3, the flux difference is $(c_k\mu)(\eta_\Lambda = \sigma) - (c_k\mu)(\eta_\Lambda = \sigma^k)$. Hence the entropy production (4.1) is at most

$$\frac{K}{2} \sum_{k\in\partial\Lambda} \sum_{\sigma} \big|(c_k\mu)(\eta_\Lambda = \sigma) - (c_k\mu)(\eta_\Lambda = \sigma^k)\big| = \frac{K}{2} \sum_{k\in\partial\Lambda} \big\|c_k\mu - (c_k\mu)\circ\theta_k^{-1}\big\|_\Lambda.$$

By (4.1), multiplying by 2 gives the claim. □

Remark 4.3. Behind Proposition 4.2 is the fact that the stationary probability current of the finite-volume dynamics can only be driven by flips at the boundary. This can also be seen from a cycle decomposition. Decompose the stationary current into cycles of the hypercube $\{-1,1\}^\Lambda$ that are traversed in the direction of the current. By (3.2), the current through an interior transition $\sigma \to \sigma^k$ is $a\,\big(h(\sigma) - h(\sigma^k)\big)$, where $h(\sigma) = \mu(\eta_\Lambda = \sigma)\exp\big(\sum_{F\subseteq\Lambda} J_F\chi_F(\sigma)\big)$ and $a > 0$ does not depend on the direction. Since h strictly decreases along such transitions, no cycle can consist of interior transitions only. Every cycle therefore passes through $\partial\Lambda$.

Remark 4.4. By Proposition 4.2, Pinsker's inequality (H3) and the Cauchy–Schwarz inequality, the left-hand side of Proposition 4.2 is at most $K^2c_+|\partial\Lambda|$. This bound alone does not show that it vanishes. To prove that, one has to combine the estimates for many different sets Λ.

5. The capacity bound

For finite $\Lambda \subset \mathbb{Z}^d$ let

$$\mathrm{cap}(\Lambda) := \inf\Big\{ \sum_{\{k,l\}\colon 0<|k-l|<L} \big(\psi(k) - \psi(l)\big)^2 :\ \psi : \mathbb{Z}^d \to \mathbb{R} \text{ finitely supported},\ \psi \ge 1 \text{ on } \Lambda \Big\},$$

where the sum runs over unordered pairs. This is the capacity of Λ in the electrical network on $\mathbb{Z}^d$ with unit conductances on the pairs $\{k,l\}$ with $0 < |k-l| < L$ [LP16, Ch. 2]. Replacing ψ by $\min\{1, \max\{0,\psi\}\}$ does not increase the sum. In the infimum we may therefore restrict to functions with values in $[0,1]$ that are equal to 1 on Λ.

Theorem 5.1. *Let $d \ge 1$ and let μ be stationary. Let $\Lambda \subset \mathbb{Z}^d$ be finite and non-empty, and let $\psi : \mathbb{Z}^d \to [0,1]$ be finitely supported with $\psi = 1$ on Λ. Then*

$$\sum_{k\in\Lambda} \Big[H_\Lambda\big(c_k\mu \,\big|\, (c_k\mu)\circ\theta_k^{-1}\big) + H_\Lambda\big((c_k\mu)\circ\theta_k^{-1} \,\big|\, c_k\mu\big)\Big] \le 4K^2c_+ \sum_{\{k,l\}\colon 0<|k-l|<L} \big(\psi(k) - \psi(l)\big)^2.$$

Consequently, the left-hand side is at most $4K^2c_+\,\mathrm{cap}(\Lambda)$.

Proof. For $0 \le t < 1$ let $\Lambda_t := \{k : \psi(k) > t\}$. These sets are finite and contain Λ, and they decrease as t increases. Since ψ takes only finitely many values, $t \mapsto \Lambda_t$ is piecewise constant with finitely many jumps, so all integrals over t below are finite sums. For k with $\psi(k) > 0$ and $0 \le t < \psi(k)$, so that $k \in \Lambda_t$, put

$$a_k(t) := H_{\Lambda_t}\big(c_k\mu \,\big|\, (c_k\mu)\circ\theta_k^{-1}\big) + H_{\Lambda_t}\big((c_k\mu)\circ\theta_k^{-1} \,\big|\, c_k\mu\big), \qquad A_k := \int_0^{\psi(k)} a_k(t)\,2t\,\mathrm{d}t.$$

By (H2), a_k is non-increasing in t. Let $\delta(k) := \max\{\psi(k) - \psi(l) : |l-k| < L\} \ge 0$.

Step 1. We have $k \in \Lambda_t$ if and only if $t < \psi(k)$. Moreover, $k \in \partial\Lambda_t$ if and only if $t < \psi(k)$ and $\psi(l) \le t$ for some l with $|l-k| < L$, that is, if and only if $t \in I_k := [\psi(k) - \delta(k), \psi(k))$.

Step 2. Let $S := \int_0^1 \sum_{k\in\Lambda_t} a_k(t)\,2t\,\mathrm{d}t = \sum_{k\colon\psi(k)>0} A_k$. Since $\Lambda \subseteq \Lambda_t$ for all $t < 1$, (H2) and (H1) show that the left-hand side of the theorem is at most $\sum_{k\in\Lambda_t} a_k(t)$ for every t. Since $\int_0^1 2t\,\mathrm{d}t = 1$, it is therefore at most S.

Step 3. By (H3), applied to both relative entropies, $\|c_k\mu - (c_k\mu)\circ\theta_k^{-1}\|_{\Lambda_t}^2 \le \mu(c_k)\,a_k(t) \le c_+\,a_k(t)$. Proposition 4.2 for Λ_t and Step 1 give

$$S \le K\sqrt{c_+}\int_0^1 \sum_{k\in\partial\Lambda_t} a_k(t)^{1/2}\,2t\,\mathrm{d}t = K\sqrt{c_+}\sum_{k:\,\delta(k)>0}\int_{I_k} a_k(t)^{1/2}\,2t\,\mathrm{d}t.$$

Step 4. Fix k with $\delta(k)>0$, and let ν be the measure $2t\,\mathrm{d}t$ on $[0,\psi(k))$, so that $\nu([0,\psi(k))) = \psi(k)^2$. Since a_k is non-increasing and I_k is a final segment of $[0,\psi(k))$, the ν-average of a_k over I_k is at most its ν-average over $[0,\psi(k))$. (If $\delta(k)=\psi(k)$ there is nothing to prove. Otherwise the average over $[0,\psi(k)-\delta(k))$ is at least $a_k(\psi(k)-\delta(k))$, which is at least the average over I_k, and the average over $[0,\psi(k))$ is a convex combination of the two.) Hence $\int_{I_k} a_k\,\mathrm{d}\nu \le \nu(I_k)A_k/\psi(k)^2$. By the Cauchy–Schwarz inequality and $\nu(I_k) = \psi(k)^2 - (\psi(k)-\delta(k))^2 \le 2\psi(k)\delta(k)$,

$$\int_{I_k} a_k^{1/2}\,\mathrm{d}\nu \le \nu(I_k)^{1/2}\Big(\int_{I_k} a_k\,\mathrm{d}\nu\Big)^{1/2} \le \frac{\nu(I_k)}{\psi(k)}\,A_k^{1/2} \le 2\,\delta(k)\,A_k^{1/2}.$$

Step 5. By Steps 3 and 4 and the Cauchy–Schwarz inequality,

$$S \le 2K\sqrt{c_+}\sum_k \delta(k)A_k^{1/2} \le 2K\sqrt{c_+}\Big(\sum_k \delta(k)^2\Big)^{1/2} S^{1/2},$$

so $S \le 4K^2c_+\sum_k \delta(k)^2$. Finally, $\delta(k)^2 \le \sum_{l:\,0<|l-k|<L}(\psi(k)-\psi(l))_+^2$. Summing over k, each unordered pair $\{k,l\}$ contributes $(\psi(k)-\psi(l))_+^2+(\psi(l)-\psi(k))_+^2 = (\psi(k)-\psi(l))^2$. Together with Step 2 this proves the first claim. The second follows from the remark before the theorem. □

Remark 5.2. The weight $2t\,\mathrm{d}t$ makes the bound quadratic in the increments of ψ. With the weight $\mathrm{d}t$, the same argument gives $\int_0^1 \sum_{k\in\Lambda_t} a_k(t)\,\mathrm{d}t \le K^2c_+\sum_{k:\,\psi(k)>0}\delta(k)^2/\psi(k)$, and the factor $1/\psi(k)$ is large where ψ is small. Integrating with $2t\,\mathrm{d}t$ over the level sets of ψ is the same as integrating with $\mathrm{d}s$ over the level sets of ψ^2.

6. Proof of Theorem 1.1

Lemma 6.1. *If $d\in\{1,2\}$, then $\mathrm{cap}(\Lambda)=0$ for every finite $\Lambda\subset\mathbb{Z}^d$.*

Proof. If $L=1$, there are no pairs $\{k,l\}$ with $0<|k-l|<L$, and there is nothing to prove. Let $L\ge 2$. Since $\mathrm{cap}(\Lambda)$ is non-decreasing in Λ, it suffices to consider $\Lambda = B_m$ with $m\ge L$. In both cases below, $\psi(k)=\rho(|k|)$ for a non-increasing function $\rho:[0,\infty)\to[0,1]$ with $\rho=1$ on $[0,m]$ and $\rho=0$ on $[n,\infty)$ for some $n>m$. To bound $\sum_{\{k,l\}}(\psi(k)-\psi(l))^2$, we assign each pair $\{k,l\}$ with $0<|k-l|<L$ and $\psi(k)\ne\psi(l)$ to an endpoint of larger norm, say k. Then $|l|\le|k|\le|l|+L-1$, and $\psi(k)\ne\psi(l)$ implies $|k|>m$ and $|l|<n$, so that $m<|k|<n+L-1$.

$d=1$. Let $\rho(r) := \big(1-(r-m)_+/(n-m)\big)_+$. Then ρ is $\frac{1}{n-m}$-Lipschitz, so every pair contributes at most $(L-1)^2/(n-m)^2$. There are at most $2(n+L)$ sites k with $m<|k|<n+L-1$, and each has $2(L-1)$ sites l with $0<|k-l|<L$. Hence

$$\sum_{\{k,l\}:\,0<|k-l|<L}\big(\psi(k)-\psi(l)\big)^2 \le \frac{4(n+L)(L-1)^3}{(n-m)^2} \xrightarrow[n\to\infty]{} 0.$$

$d=2$. Let $\rho(r) := \min\big\{1,\max\{0,\log(n/r)/\log(n/m)\}\big\}$ for $r>0$ and $\rho(0):=1$. For $0<r\le r'$,

$$0 \le \rho(r)-\rho(r') \le \frac{r'-r}{\max(r,m)\log(n/m)},$$

because $|\rho'(u)| \le \mathbf{1}_{(m,n)}(u)/(u\log(n/m))$. Consider a pair as above. Then $\max(|l|,m) \ge |k|/2$: either $|l| \ge |k|/2$, or $|k|/2 < |k| - |l| \le L-1 < m$. Hence

$$\big(\psi(k)-\psi(l)\big)^2 \le \frac{(L-1)^2}{\max(|l|,m)^2\log^2(n/m)} \le \frac{4(L-1)^2}{|k|^2\log^2(n/m)}.$$

Since $\#\{k\in\mathbb{Z}^2 : |k| = r\} = 8r$ for $r \ge 1$, and each k has at most $(2L-1)^2$ sites l with $0 < |k-l| < L$,

$$\begin{aligned}\sum_{\{k,l\}:\,0<|k-l|<L} \big(\psi(k)-\psi(l)\big)^2 &\le \sum_{r=m+1}^{n+L} 8r\,(2L-1)^2\,\frac{4(L-1)^2}{r^2\log^2(n/m)} \\ &\le \frac{32(2L-1)^2(L-1)^2\,\log\big((n+L)/m\big)}{\log^2(n/m)}.\end{aligned}$$

Here we used $\sum_{r=m+1}^{M} 1/r \le \log(M/m)$. The right-hand side tends to 0 as $n\to\infty$. □

Proof of Theorem 1.1. The "if" part is Corollary 3.4. For the "only if" part, let $d \in \{1,2\}$ and let μ be stationary. By Remark 2.2, this includes measures that are stationary for the semigroup. By Theorem 5.1 and Lemma 6.1, the left-hand side of (4.2) vanishes for every $n \ge 0$. Hence μ is a Gibbs state for J. □

In terms of Section 3, the proof shows that in $d \le 2$ the finite-volume dynamics with random boundary condition is reversible in every box.

7. Conservative dynamics

In this section the spin flips are replaced by exchanges of neighbouring spins. Let $\mathbb{B}$ be the set of nearest-neighbour bonds $b = \{x,y\}$, $|x-y|_1 = 1$. For such a bond, η^{xy} is obtained from η by exchanging η_x and η_y, and $\theta_b : E \to E$ is the map $\eta \mapsto \eta^{xy}$; similarly σ^{xy} for $\sigma \in \{-1,1\}^\Lambda$ and $x,y \in \Lambda$. We write $\operatorname{dist}(z,b) := \min\{|z-x|,|z-y|\}$.

Setting. Let J be a potential with range L as in Section 2. Exchange rates $(c_b)_{b\in\mathbb{B}}$ are *associated with* J if

$$c_b(\eta) = b_b(\eta)\exp\Big(\frac12\sum_F J_F\big(\chi_F(\eta)-\chi_F(\eta^{xy})\big)\Big),$$

where $b_b : E \to (0,\infty)$ depends only on the coordinates η_z with $\operatorname{dist}(z,b) < L$, satisfies $b_b(\eta^{xy}) = b_b(\eta)$, and takes values in $[b_-,b_+] \subset (0,\infty)$. Only sets F containing exactly one of x,y contribute to the sum, so $c_- \le c_b \le c_+$ with $c_\pm := b_\pm e^{\pm 2\|J\|}$, and

$$\log\frac{c_b(\eta)}{c_b(\eta^{xy})} = \sum_F J_F\big(\chi_F(\eta)-\chi_F(\eta^{xy})\big). \tag{7.1}$$

The generator is $\mathcal{L}f(\eta) := \sum_{b\in\mathbb{B}} c_b(\eta)\big(f(\eta^{xy})-f(\eta)\big)$ for local f. Stationarity is defined as in Definition 2.1, and as in (2.4),

$$\int \mathcal{L}f\,\mathrm{d}\mu = \sum_{b\in\mathbb{B}}\Big[\int f\,\mathrm{d}\big((c_b\mu)\circ\theta_b^{-1}\big) - \int f\,\mathrm{d}(c_b\mu)\Big]. \tag{7.2}$$

Definition 7.1. A probability measure μ is *reversible* for the exchange dynamics if $(c_b\mu)\circ\theta_b^{-1} = c_b\mu$ for every $b \in \mathbb{B}$.

As in Lemma 3.3, μ is reversible if and only if, for every b, the measure $\exp\big(\frac{1}{2}\sum_F J_F(\chi_F - \chi_F\circ\theta_b)\big)\mu$ is θ_b-invariant; in particular this does not depend on the choice of the b_b. Reversible measures are stationary by (7.2). Every Gibbs state for J, and for J plus a constant magnetic field, is reversible. More generally, the reversible measures are the canonical Gibbs measures for J, cf. [Ge79]. They are the conservative analogue of Gibbs states.

Definition 7.2. A probability measure μ has *finite energy* if there is $\delta > 0$ such that

$$\mu(\eta_y = s \,|\, \eta_\Lambda = \sigma) \geq \delta$$

for every finite Λ, every $y \notin \Lambda$, every $s \in \{-1,1\}$ and every $\sigma \in \{-1,1\}^\Lambda$ with $\mu(\eta_\Lambda = \sigma) > 0$.

If μ has finite energy, then $\mu(\eta_\Lambda = \sigma) \geq \delta^{|\Lambda|} > 0$ for all Λ and σ. Gibbs states for J, with or without a constant magnetic field, have finite energy.

Theorem 7.3. *Let $d \in \{1,2\}$. Every stationary measure of the exchange dynamics that has finite energy is reversible.*

For spin flips no assumption on μ was needed, because the rate of a flip at a site of Λ lies in $[c_-, c_+]$ whatever the configuration outside Λ. An exchange across the boundary of Λ, in contrast, changes η_Λ only if the outer spin is opposite to the inner one, so its effective rate involves the conditional law of the outer spin. Finite energy keeps this rate away from zero. Remark 7.6 discusses the assumption.

Finite-volume dynamics and the boundary estimate. Let μ have finite energy with constant δ, and let V be finite and non-empty. For a bond $b = \{x, y\}$ meeting V let

$$\nu_b := \mathbf{1}_{\{\eta_x \neq \eta_y\}}\, c_b\mu, \qquad a_b(V) := H_V\big(\nu_b \,\big|\, \nu_b\circ\theta_b^{-1}\big) + H_V\big(\nu_b\circ\theta_b^{-1} \,\big|\, \nu_b\big).$$

Since θ_b is the identity on $\{\eta_x = \eta_y\}$, we have $(c_b\mu)\circ\theta_b^{-1} - c_b\mu = \nu_b\circ\theta_b^{-1} - \nu_b$. For $b \subseteq V$, the atoms $\{\eta_V = \sigma\}$ with $\sigma_x = \sigma_y$ are fixed by θ_b, so

$$a_b(V) = H_V\big(c_b\mu \,\big|\, (c_b\mu)\circ\theta_b^{-1}\big) + H_V\big((c_b\mu)\circ\theta_b^{-1} \,\big|\, c_b\mu\big) \qquad (b \subseteq V).$$

We call a bond b *interior* to V if $\{z : \mathrm{dist}(z, b) < L\} \subseteq V$, and we let $B(V)$ be the set of bonds that meet V and are not interior to V.

Proposition 7.4. *Let μ be stationary with finite energy, and let V be finite and non-empty. Then*

$$\sum_{b\subseteq V} a_b(V) \leq K_\delta \sum_{b\in B(V)} \big\|\nu_b - \nu_b\circ\theta_b^{-1}\big\|_V, \qquad K_\delta := \log\frac{2d\,c_+}{\delta\,c_-} + 4\|J\|.$$

Proof. Balance equations. Apply (7.2) to $f = \mathbf{1}_{\{\eta_V = \sigma\}}$. Only bonds meeting V contribute, and we obtain

$$\sum_{b\cap V\neq\emptyset} \Big[\big(\nu_b\circ\theta_b^{-1}\big)(\eta_V = \sigma) - \nu_b(\eta_V = \sigma)\Big] = 0. \tag{7.3}$$

For $b \subseteq V$ we have $\big(\nu_b\circ\theta_b^{-1}\big)(\eta_V = \sigma) = \nu_b(\eta_V = \sigma^{xy})$. For a bond $b = \{x, y\}$ with $x \in V$ and $y \notin V$, the map θ_b sends $\{\eta_V = \sigma^x,\ \eta_x \neq \eta_y\}$ onto $\{\eta_V = \sigma,\ \eta_x \neq \eta_y\}$, so $\big(\nu_b\circ\theta_b^{-1}\big)(\eta_V = \sigma) = \nu_b(\eta_V = \sigma^x)$. Hence (7.3) says that the law of η_V under μ is stationary for the following Markov chain on $\{-1,1\}^V$, the finite-volume dynamics with random boundary condition:

- for each bond $b \subseteq V$, the chain jumps from σ to σ^{xy} with stationary flux $\nu_b(\eta_V = \sigma)$, i.e. at rate $\mu(c_b \,|\, \eta_V = \sigma) \in [c_-, c_+]$ if $\sigma_x \neq \sigma_y$;

- for each $x \in V$ with a neighbour outside V, the chain jumps from σ to σ^x with stationary flux $\sum_y \nu_{\{x,y\}}(\eta_V = \sigma)$, where the sum runs over the neighbours $y \notin V$ of x. The corresponding rate is $\sum_y \mu\big(c_{\{x,y\}}\mathbf{1}_{\{\eta_y=-\sigma_x\}} \,\big|\, \eta_V = \sigma\big)$, and it lies in $[\delta c_-, 2d\,c_+]$ by finite energy.

The flux in the reverse direction of a transition caused by b is $\big(\nu_b \circ \theta_b^{-1}\big)(\eta_V = \sigma)$.

Entropy production. Apply Lemma 4.1 to this chain with $W(\sigma) := \sum_{F\subseteq V} J_F \chi_F(\sigma)$ in place of V there. Transitions caused by different bonds $b \subseteq V$ are different, so the entropy production of the chain is at least $\frac{1}{2}\sum_{b\subseteq V} a_b(V)$; the transitions $\sigma \to \sigma^x$ contribute non-negative terms, which we drop. We now bound the bracket in Lemma 4.1.

- If b is interior to V, then c_b depends only on η_V, and every F with $J_F \neq 0$ that meets b is contained in V. By (7.1) the bracket vanishes.
- For the other bonds $b \subseteq V$, the bracket is at most $\log(c_+/c_-) + |W(\sigma) - W(\sigma^{xy})| \leq \log(c_+/c_-) + 4\|J\|$ in absolute value.
- For the transitions $\sigma \to \sigma^x$, it is at most $\log \frac{2d\,c_+}{\delta c_-} + 2\|J\|$ in absolute value.

All brackets are bounded by K_δ, and the difference of the forward and backward fluxes of $\sigma \to \sigma^x$ is at most $\sum_y \big|\nu_{\{x,y\}}(\eta_V = \sigma) - \nu_{\{x,y\}}(\eta_V = \sigma^x)\big|$. Hence the entropy production of the chain is at most $\frac{K_\delta}{2}\sum_{b\in B(V)} \|\nu_b - \nu_b \circ \theta_b^{-1}\|_V$. □

The capacity bound. For finitely supported $\psi : \mathbb{Z}^d \to \mathbb{R}$ we use the nearest-neighbour Dirichlet energy $\sum_{b=\{k,l\}\in\mathbb{B}} (\psi(k) - \psi(l))^2$.

Theorem 7.5. *Let $d \geq 1$, let μ be stationary with finite energy, let Λ be finite and non-empty, and let $\psi : \mathbb{Z}^d \to [0,1]$ be finitely supported with $\psi = 1$ on Λ. Then*

$$\sum_{b\subseteq\Lambda} \Big[H_\Lambda\big(c_b\mu \,\big|\, (c_b\mu)\circ\theta_b^{-1}\big) + H_\Lambda\big((c_b\mu)\circ\theta_b^{-1} \,\big|\, c_b\mu\big)\Big] \leq C \sum_{\{k,l\}\in\mathbb{B}} \big(\psi(k) - \psi(l)\big)^2,$$

where C depends only on d, L, $c_\pm$, $\|J\|$ and δ.

Proof. For $0 \leq t < 1$ let $\Lambda_t := \{\psi > t\} \supseteq \Lambda$. For a bond $b = \{x, y\}$ put

$$\psi_+(b) := \max\{\psi(x), \psi(y)\}, \qquad \psi_-(b) := \min\{\psi(x), \psi(y)\},$$
$$\underline{\psi}(b) := \min\{\psi(z) : \operatorname{dist}(z, b) < L\},$$

and $\delta(b) := \psi_+(b) - \underline{\psi}(b) \geq 0$. Then b meets Λ_t if and only if $t < \psi_+(b)$, $b \subseteq \Lambda_t$ if and only if $t < \psi_-(b)$, and $b \in B(\Lambda_t)$ if and only if $t \in I_b := [\underline{\psi}(b), \psi_+(b))$. For $t < \psi_+(b)$ let $a_b(t) := a_b(\Lambda_t)$. By (H2), a_b is non-increasing. Let ν be the measure $2t\,\mathrm{d}t$, put $A_b := \int_0^{\psi_-(b)} a_b\,\mathrm{d}\nu$, and let $S := \int_0^1 \sum_{b\subseteq\Lambda_t} a_b(t)\,\nu(\mathrm{d}t) = \sum_b A_b$. As in Step 2 of the proof of Theorem 5.1, the left-hand side of the claim is at most S.

By Proposition 7.4 for Λ_t, integrated against ν,

$$S \leq K_\delta \sum_b \int_{I_b} \big\|\nu_b - \nu_b\circ\theta_b^{-1}\big\|_{\Lambda_t}\,\nu(\mathrm{d}t).$$

Call b *good* if $\delta(b) > 0$ and $\underline{\psi}(b) \geq \psi_+(b)/2$, and *bad* if $\delta(b) > 0$ and $\underline{\psi}(b) < \psi_+(b)/2$.

- For a bad bond we use $\|\nu_b - \nu_b\circ\theta_b^{-1}\| \leq 2\nu_b(E) \leq 2c_+$. Since $\psi_+(b) < 2\delta(b)$, its contribution is at most $2c_+K_\delta\,\nu(I_b) \leq 2c_+K_\delta\,\psi_+(b)^2 \leq 8c_+K_\delta\,\delta(b)^2$.
- For a good bond, Pinsker's inequality (H3) gives $\|\nu_b - \nu_b\circ\theta_b^{-1}\|_{\Lambda_t} \leq (c_+ a_b(t))^{1/2}$. Since a_b is non-increasing and $\underline{\psi}(b) \leq \psi_-(b)$, we have $\int_{I_b} a_b\,\mathrm{d}\nu \leq \nu(I_b)\,a_b(\underline{\psi}(b))$ and $A_b \geq$

$\underline{\psi}(b)^2 a_b(\underline{\psi}(b))$. Together with the Cauchy–Schwarz inequality and $\nu(I_b) = \psi_+(b)^2 - \underline{\psi}(b)^2 \le 2\psi_+(b)\delta(b)$, this gives

$$\int_{I_b} a_b^{1/2}\,\mathrm{d}\nu \le \nu(I_b)^{1/2}\Big(\frac{\nu(I_b)}{\underline{\psi}(b)^2}A_b\Big)^{1/2} \le \frac{2\psi_+(b)\delta(b)}{\underline{\psi}(b)}A_b^{1/2} \le 4\,\delta(b)\,A_b^{1/2}.$$

With $D := \sum_b \delta(b)^2$ and the Cauchy–Schwarz inequality we obtain $S \le 4K_\delta\sqrt{c_+ D}\,S^{1/2} + 8c_+K_\delta D$. Hence $S \le 16K_\delta(K_\delta+1)c_+D$, because $S \le aS^{1/2}+c$ implies $S \le a^2+2c$.

Finally, $\delta(b) \le \max\{|\psi(z)-\psi(z')| : z, z' \in Q(b)\}$, where $Q(b)$ is the cube of side length $2L+3$ centred at one endpoint of b. Any two points of $Q(b)$ are joined by a nearest-neighbour path in $Q(b)$ of length at most $d(2L+2)$. By the Cauchy–Schwarz inequality, $\delta(b)^2 \le d(2L+2)\sum_{e\subseteq Q(b)}|\psi(k_e)-\psi(l_e)|^2$, where $e=\{k_e,l_e\}$ runs over the nearest-neighbour bonds in $Q(b)$. Every such bond lies in $Q(b)$ for at most $2d(2L+3)^d$ bonds b. Therefore $D \le 2d^2(2L+2)(2L+3)^d\sum_{\{k,l\}\in\mathbb{B}}(\psi(k)-\psi(l))^2$. □

Proof of Theorem 7.3. Let $d \le 2$. The proof of Lemma 6.1, applied with $L=2$, provides cut-offs ψ with $\psi = 1$ on B_n and arbitrarily small nearest-neighbour Dirichlet energy, since nearest neighbours satisfy $|k-l|<2$. By Theorem 7.5 and (H1), $c_b\mu$ and $(c_b\mu)\circ\theta_b^{-1}$ agree on $\mathcal{F}_{B_n}$ for all n and all $b \subseteq B_n$. As in the proof of Lemma 3.3, the π-λ theorem gives $(c_b\mu)\circ\theta_b^{-1} = c_b\mu$ for every b. □

Remark 7.6 (On finite energy). (a) Finite energy is used only in Proposition 7.4, to bound the rates of the boundary transitions from below, and to make all marginals strictly positive. Without it the constant K_δ is infinite, and we do not know whether Theorem 7.3 remains true for interacting exchange dynamics. Some stationary measures do violate finite energy; an example is the point mass on $\eta \equiv +1$, which is reversible.

(b) For the symmetric simple exclusion process ($J=0$ and b_b constant) the assumption can be dropped. By Liggett's characterisation [Li73, Li74] (see also [Li85, Ch. VIII]), the extremal stationary measures of symmetric exclusion with symmetric irreducible jump rates are indexed by the harmonic functions of one random walk with values in $[0,1]$. On a recurrent graph these are constant, so every stationary measure is a mixture of product Bernoulli measures, hence exchangeable and reversible.

Remark 7.7 (Recurrent graphs). As in Remark 9.3, Proposition 7.4 and Theorem 7.5 only use that the bonds form a connected graph of bounded degree. On every graph of bounded degree on which simple random walk is recurrent, every stationary measure of the exchange dynamics that has finite energy is reversible.

8. Non-reversible stationary measures for exchange dynamics in three dimensions

The following example shows that on a transient graph, stationary measures of exchange dynamics need not be reversible, even if they have finite energy. In the example the rates vanish on some bonds of $\mathbb{Z}^3$.

Let $d=3$, $H_+ := \{x \in \mathbb{Z}^3 : x_1 \ge 1\}$ and $H_- := \{x \in \mathbb{Z}^3 : x_1 \le 0\}$, and let $o_- := (0,0,0)$, $o_+ := (1,0,0)$. Let G be the graph with vertex set $\mathbb{Z}^3$ whose edges are the nearest-neighbour bonds with both endpoints in H_+ or both in H_-, together with the single bond $\{o_-, o_+\}$. Thus G consists of two half-spaces joined by one bridge. The *symmetric simple exclusion process on* G is the exchange dynamics with $J=0$, rate $c_b = 1$ for the edges b of G, and rate $c_b = 0$ for the remaining bonds of $\mathbb{Z}^3$. It is a Feller process whose generator acts on local functions as in Section 7 [Li85, Ch. I and VIII]. Since the rates are constant on the edges of G, a measure is

reversible if and only if it is invariant under θ_b for every edge b of G. All product Bernoulli measures with constant density are reversible.

Theorem 8.1. *Let $0 < \rho_- < \rho_+ < 1$. The symmetric simple exclusion process on G has a stationary measure that is not reversible and has finite energy with constant $\delta = \min\{\rho_-, 1-\rho_+\}$.*

Proof. Step 1: a non-constant bounded harmonic function. Let (X_t) be the continuous-time simple random walk on G, which jumps along each edge at rate 1.

- *Transience.* Fold $\mathbb{Z}^3$ along the plane $\{x_1 = \frac{1}{2}\}$, i.e. map x to x if $x_1 \geq 1$ and to $(1 - x_1, x_2, x_3)$ if $x_1 \leq 0$. By the reflection symmetry of simple random walk on $\mathbb{Z}^3$, the image of that walk is a Markov chain on H_+. It coincides with simple random walk on H_+, except for additional jumps from a site to itself at the sites with $x_1 = 1$. Since simple random walk on $\mathbb{Z}^3$ is transient [LP16, Ch. 2], so is simple random walk on H_+. By Rayleigh's monotonicity principle [LP16, Ch. 2], $G \supseteq H_+$ is transient as well.
- *The function α.* A transient walk crosses the bridge $\{o_-, o_+\}$ only finitely often, so almost surely it eventually stays in H_+ or eventually stays in H_-. Let
$$\alpha(x) := P_x(X \text{ eventually stays in } H_+).$$
By the Markov property, α is harmonic on G, that is $\sum_{y\sim x}(\alpha(y) - \alpha(x)) = 0$ for every x, where $y \sim x$ means that $\{x, y\}$ is an edge of G. Moreover $0 \leq \alpha \leq 1$.
- *α is not constant.* The reflection $x \mapsto (1 - x_1, x_2, x_3)$ is an automorphism of G exchanging H_+ and H_-, so $\alpha(1 - x_1, x_2, x_3) = 1 - \alpha(x)$. A constant α would therefore equal $\frac{1}{2}$. Let $h(x)$ be the probability that the walk started at $x \in H_+$ ever visits o_+. Before visiting o_+, it moves as simple random walk Y on H_+. For Y started at o_+, the Markov property gives
$$E[h(Y_t)] = P(Y \text{ visits } o_+ \text{ after time } t),$$
and this tends to 0 because Y is transient. Hence $h(x) < \frac{1}{2}$ for some $x \in H_+$. A walk from this x that never visits o_+ never uses the bridge and stays in H_+, so $\alpha(x) \geq 1 - h(x) > \frac{1}{2}$.

Put $\beta := \rho_- + (\rho_+ - \rho_-)\alpha$. Then β is harmonic on G, not constant, and takes values in $[\rho_-, \rho_+]$.

Step 2: the density is preserved. Let ν_β be the product measure on E with $\nu_\beta(\eta_x = +1) = \beta(x)$, and let $\nu_\beta T_t$ be the law at time t of the exclusion process started from ν_β. Since $\mathcal{L}\mathbf{1}_{\{\eta_x=+1\}} = \sum_{y\sim x}\big(\mathbf{1}_{\{\eta_y=+1\}} - \mathbf{1}_{\{\eta_x=+1\}}\big)$, the function $u_t(x) := (\nu_\beta T_t)(\eta_x = +1)$ satisfies $\frac{\mathrm{d}}{\mathrm{d}t}u_t(x) = \sum_{y\sim x}(u_t(y) - u_t(x))$, with $u_0 = \beta$. Put $w_t := u_t - \beta$. Since β is harmonic, w solves the same equation with $w_0 = 0$. Every vertex of G has degree at most 6, so $\varphi(t) := \sup_x |w_t(x)| \leq 1$ satisfies $\varphi(t) \leq 12 \int_0^t \varphi(s)\,\mathrm{d}s$, and therefore $\varphi \equiv 0$. Hence $(\nu_\beta T_t)(\eta_x = +1) = \beta(x)$ for all $t \geq 0$ and $x \in \mathbb{Z}^3$.

Step 3: a stationary limit. The space of probability measures on E is weakly compact. Choose $T_n \to \infty$ such that $\frac{1}{T_n}\int_0^{T_n} \nu_\beta T_t\,\mathrm{d}t$ converges weakly to some μ^*. For continuous f and $s \geq 0$,
$$\int T_s f\,\mathrm{d}\mu^* - \int f\,\mathrm{d}\mu^* = \lim_{n\to\infty} \frac{1}{T_n}\Big(\int_{T_n}^{T_n+s} - \int_0^s\Big) \int f\,\mathrm{d}(\nu_\beta T_t)\,\mathrm{d}t = 0,$$
because $T_s f$ is continuous. Hence μ^* is invariant, and in particular stationary in the sense of Definition 2.1. By Step 2, $\mu^*(\eta_x = +1) = \beta(x)$ for every x.

Step 4: μ^* is not reversible. If μ^* were reversible, it would be invariant under θ_b for every edge $b = \{x, y\}$ of G, and so $\beta(x) = \mu^*(\eta_x = +1) = \mu^*(\eta_y = +1) = \beta(y)$. Since G is connected, β would be constant, contradicting Step 1.

Step 5: finite energy. We use the stirring construction of the exclusion process [Li85, Ch. VIII]. Attach independent Poisson clocks of rate 1 to the edges of G, independent of

the initial configuration, and exchange the spins at x and y whenever the clock of $\{x,y\}$ rings. Since G has bounded degree, this defines for every t a random bijection π_t of $\mathbb{Z}^3$, independent of η_0, such that $\eta_t(\pi_t(z)) = \eta_0(z)$ for all z, and (η_t) is the exclusion process on G. Let η_0 have law ν_β. Conditionally on π_t, the spins $\eta_t(x) = \eta_0(\pi_t^{-1}(x))$, $x \in \mathbb{Z}^3$, are independent, and $\eta_t(x) = +1$ with conditional probability $\beta(\pi_t^{-1}(x)) \in [\rho_-, \rho_+]$. Hence, for finite Λ, $y \notin \Lambda$ and $\sigma \in \{-1,1\}^\Lambda$,

$$\rho_- \, (\nu_\beta T_t)(\eta_\Lambda = \sigma) \le (\nu_\beta T_t)(\eta_\Lambda = \sigma,\ \eta_y = +1) \le \rho_+ \, (\nu_\beta T_t)(\eta_\Lambda = \sigma),$$

the same holds for $\eta_y = -1$ with the constants $1-\rho_+$ and $1-\rho_-$, and $(\nu_\beta T_t)(\eta_\Lambda = \sigma) \ge \delta^{|\Lambda|}$ with $\delta := \min\{\rho_-, 1-\rho_+\}$. These inequalities survive averaging over $t \in [0, T_n]$ and the weak limit, because cylinder events are open and closed. Hence $\mu^*(\eta_\Lambda = \sigma) > 0$ and $\mu^*(\eta_y = s \,|\, \eta_\Lambda = \sigma) \ge \delta$ for all finite Λ, $y \notin \Lambda$, $s \in \{-1,1\}$ and σ. □

Remark 8.2. (a) The measure μ^* is the measure μ_β of Liggett's characterisation [Li73, Li85]; in fact $\nu_\beta T_t$ converges as $t \to \infty$. Step 5 shows that μ^* is a mixture of product Bernoulli measures with densities in $[\rho_-, \rho_+]$; it is not reversible because these densities are not constant in space.

(b) The restriction $0 < \rho_- < \rho_+ < 1$ is needed for finite energy. For $\rho_- = 0$ and $\rho_+ = 1$, i.e. $\beta = \alpha$, the construction still gives a stationary measure that is not reversible, but $\alpha(x) \to 0$ as $|x| \to \infty$ in H_-. Indeed, for $x \in H_-$ the walk must visit o_- before it can end up in H_+, and before that visit it moves as simple random walk on H_-. By the folding of Step 1, applied to H_-, this walk is the image of simple random walk on $\mathbb{Z}^3$, and o_- has the preimages o_- and o_+. Hence $\alpha(x) \le P_x(\text{simple random walk on } \mathbb{Z}^3 \text{ visits } o_- \text{ or } o_+)$. This tends to 0, because the Green's function of $\mathbb{Z}^3$ tends to 0 at infinity. (By dominated convergence: the heat kernel satisfies $p_t(x,z) \le p_t(z,z)$ and $\int_0^\infty p_t(z,z)\,\mathrm{d}t < \infty$.) Since $\mu^*(\eta_x = +1) = \alpha(x)$ and finite energy implies $\mu^*(\eta_x = +1) \ge \delta$, finite energy fails.

(c) The graph G is connected and of bounded degree, the rates are positive on all its edges, and μ^* has finite energy. Hence the recurrence assumption in the graph version of Theorem 7.3 (Remark 7.7) cannot be dropped.

(d) The example illustrates the role of transience. The density β is a non-constant bounded harmonic function, and in the stationary state there is a particle current through the bridge, fed by the two ends of G, where the densities differ. On a recurrent graph, bounded harmonic functions are constant. This follows from the linear version of Theorem 5.1: for harmonic h and ψ as there,

$$\sum_{\{k,l\}} \min\{\psi(k), \psi(l)\}^2 \big(h(k) - h(l)\big)^2 \le C \, \|h\|_\infty^2 \sum_{\{k,l\}} \big(\psi(k) - \psi(l)\big)^2.$$

Here C is a universal constant. This is the classical Caccioppoli inequality, obtained by testing harmonicity with $h\psi^2$, and on a recurrent graph the right-hand side can be made arbitrarily small. If $H_\pm$ are replaced by half-planes of $\mathbb{Z}^2$, which are recurrent, the construction fails, in accordance with Theorem 7.3 and Remark 7.6(b).

(e) The example uses vanishing rates on all bonds crossing the plane $\{x_1 = \frac{1}{2}\}$ except one. Suppose instead that every nearest-neighbour bond of $\mathbb{Z}^3$ carries a rate in $[c_-, c_+] \subset (0, \infty)$ that does not depend on the configuration. Then bounded harmonic functions of the walk are constant by the elliptic Harnack inequality [De99]. By Liggett's characterisation, every stationary measure of the corresponding exclusion process is then reversible, in every dimension. For interacting exchange dynamics with rates bounded away from zero on all bonds of $\mathbb{Z}^d$, $d \ge 3$, we do not know whether non-reversible stationary measures exist. As for spin flips (Remark 9.2), the capacity bound only shows that the entropy production in B_n of such a measure with finite energy would be $O(n^{d-2})$.

Interacting exchange dynamics. The example is not tied to the absence of interaction. We now transfer it to exchange dynamics with summable interactions, by a coupling argument in the spirit of Lammers and Toninelli [LT24]; see Remark 8.4(a).

Let J be a potential with range L such that

$$\sum_F |J_F| < \infty, \tag{8.1}$$

and put $j(z) := \sum_{F \ni z} |J_F|$ for $z \in \mathbb{Z}^3$. For instance, (8.1) holds if only finitely many J_F are non-zero, whatever their size. For an edge $b = \{x, y\}$ of G let

$$c_b(\eta) := \exp\Big(\frac{1}{2}\sum_F J_F\big(\chi_F(\eta) - \chi_F(\eta^{xy})\big)\Big), \tag{8.2}$$

and let $c_b := 0$ for the bonds of $\mathbb{Z}^3$ that are not edges of G. On the edges of G these are the rates of Section 7 with $b_b = 1$; as for the exclusion process above, they vanish on the other bonds. Only sets F that contain exactly one of x and y contribute to (8.2), so

$$\varepsilon_b := \sup_\eta |c_b(\eta) - 1| \le e^{j(x)+j(y)} - 1.$$

By (8.1), $\Phi := \sum_F J_F \chi_F$ is a bounded continuous function on E, and (7.1) becomes

$$\log \frac{c_b(\eta)}{c_b(\eta^{xy})} = \Phi(\eta) - \Phi(\eta^{xy}). \tag{8.3}$$

For $0 < \rho < 1$ let ν_ρ be the product Bernoulli measure with constant density ρ. Since ν_ρ is invariant under every θ_b, (8.3) shows that the probability measure proportional to $e^{-\Phi}\nu_\rho$ is reversible. It has finite energy, because Φ is bounded. Finally, let

$$g(z, x) := \int_0^\infty P_z(X_t = x)\,\mathrm{d}t \qquad (z, x \in \mathbb{Z}^3)$$

be the Green's function of the random walk X of Step 1 of the proof of Theorem 8.1, and put

$$\gamma(x) := \sum_{b=\{y,z\}} \varepsilon_b \big(g(y, x) + g(z, x)\big) \qquad (x \in \mathbb{Z}^3), \tag{8.4}$$

where the sum runs over the edges of G.

Theorem 8.3. *Assume (8.1), let $0 < \rho_- < \rho_+ < 1$, and let $\beta = \rho_- + (\rho_+ - \rho_-)\alpha$ be the function of Step 1 of the proof of Theorem 8.1. Then $\gamma(x) \to 0$ as $|x| \to \infty$, and the exchange dynamics on G with rates (8.2) has a stationary measure μ such that*

$$\big|\mu(\eta_x = +1) - \beta(x)\big| \le \gamma(x) \qquad \text{for all } x \in \mathbb{Z}^3.$$

In particular, $\mu(\eta_x = +1)$ tends to ρ_+ as $|x| \to \infty$ in H_+ and to ρ_- as $|x| \to \infty$ in H_-, and μ is not reversible.

Proof. Step 1: properties of γ and β. Let g_3 be the Green's function of simple random walk on $\mathbb{Z}^3$, and write $\bar{x} := (1 - x_1, x_2, x_3)$.

- *Boundedness of g.* By the strong Markov property, $g(z, x) = P_z(X \text{ visits } x)\, g(x, x) \le g(x, x)$. Let $x \in H_+$; the case $x \in H_-$ is symmetric. The expected number of visits of the jump chain of X to x is $\deg(x)$ times the effective resistance between x and infinity in G with unit conductances, and each visit lasts an exponential time with mean $1/\deg(x)$ [LP16, Ch. 2]. Hence $g(x, x)$ equals this resistance. By Rayleigh's monotonicity principle it is at most the effective resistance between x and infinity in H_+, which in the same way

equals the Green's function of simple random walk on H_+ at (x,x). By the folding of Step 1 of the proof of Theorem 8.1, the latter is $g_3(x,x)+g_3(x,\bar{x})$. Hence $g \le 2g_3(0,0)$.

- *Decay of g.* Fix z, and let $x \in H_+$. Since X is reversible with respect to the counting measure, $g(z,x) = g(x,z) \le P_x(X \text{ visits } z)\, g(z,z)$. Before its first visit to o_+, the walk started at x moves as simple random walk on H_+, and it cannot enter H_- without visiting o_+. So $P_x(X \text{ visits } z)$ is at most the probability that simple random walk on H_+ started at x visits o_+, plus, if $z \in H_+$, the probability that it visits z. For $w \in H_+$ the latter probability is at most $6\big(g_3(x,w)+g_3(x,\bar{w})\big)$, because $g_3(x,w)+g_3(x,\bar{w})$ is the Green's function of simple random walk on H_+ at (x,w), and each visit to w lasts on average at least $1/6$. These bounds tend to 0 as $|x| \to \infty$ (Remark 8.2(b)). Hence $g(z,x) \to 0$ as $|x| \to \infty$ in H_+, and likewise in H_-.
- *Summability.* Since $e^s - 1 \le se^s$ for $s \ge 0$, we have $\varepsilon_b \le e^{2\|J\|}\big(j(x)+j(y)\big)$ for $b = \{x,y\}$. Every vertex of G lies in at most 6 edges, and every F with $J_F \ne 0$ has at most L^3 elements. Hence $\sum_b \varepsilon_b \le 6e^{2\|J\|}\sum_z j(z) \le 6L^3 e^{2\|J\|}\sum_F |J_F| < \infty$.

By dominated convergence, $\gamma(x) \to 0$ as $|x| \to \infty$. By Remark 8.2(b) and the symmetry $\alpha(\bar{x}) = 1-\alpha(x)$, $\beta(x)$ tends to ρ_- as $|x| \to \infty$ in H_- and to ρ_+ as $|x| \to \infty$ in H_+. Finally, $\sum_z j(z) < \infty$ implies $j(z) \to 0$ as $|z| \to \infty$.

Step 2: coupling with the exclusion process. Let $\mathbb{B}_G$ be the set of edges of G, and let $m := e^{2\|J\|}$, so that $c_b \le m$ for all b. Let N be a Poisson point process on $[0,\infty) \times \mathbb{B}_G \times [0,m]$ with intensity $\mathrm{d}s \otimes (\text{counting measure}) \otimes \mathrm{d}u$. Thus every edge carries a Poisson clock of rate m, and every ring carries a label u that is uniformly distributed on $[0,m]$. Let η_0 have law ν_β and be independent of N, and put $\xi_0 := \eta_0$. At each point (s,b,u) of N, the process ξ exchanges the spins at the endpoints of b if $u \le 1$, and the process η does so if $u \le c_b(\eta_{s-})$. This is the usual graphical construction, which is well defined because the rates are bounded and of finite range. The process ξ is the exclusion process on G, constructed by stirring as in Step 5 of the proof of Theorem 8.1. The process η is the exchange dynamics with rates (8.2), because a clock of rate m whose rings are accepted with probability c_b/m rings at rate c_b.

For $0 \le s \le t$ let $\pi_{s,t}$ be the bijection of $\mathbb{Z}^3$ obtained, as in Step 5 of the proof of Theorem 8.1, from the transpositions along the edges b of the points (r,b,u) of N with $s < r \le t$ and $u \le 1$. For fixed s and w, the path $r \mapsto \pi_{s,r}(w)$, $r \ge s$, is the random walk X started at w, and it depends only on the points of N with first coordinate in (s,∞). Call a point (s,b,u) of N a *disagreement point* if exactly one of the inequalities $u \le 1$ and $u \le c_b(\eta_{s-})$ holds, that is, if u lies in the interval $I_b(s)$ with endpoints $\min\{1, c_b(\eta_{s-})\}$ (excluded) and $\max\{1, c_b(\eta_{s-})\}$ (included). The length $|I_b(s)|$ of this interval is at most ε_b, and $I_b(s)$ depends only on η_0 and on the points of N with first coordinate in $[0,s)$. Let

$$M_t := \big\{\pi_{s,t}(w) :\ (s,b,u) \text{ is a disagreement point with } s \le t,\ w \in b\big\}.$$

We claim that $\eta_t(x) = \xi_t(x)$ for every $x \notin M_t$. Let $z_r := \pi_{0,r}\big(\pi_{0,t}^{-1}(x)\big)$ for $0 \le r \le t$, so that $z_t = x$; this path jumps only at points (r,b,u) of N with $u \le 1$ and $z_{r-} \in b$. If there is a disagreement point (s,b,u) with $s \le t$ and $z_{s-} \in b$, then $z_s \in b$, whether or not $u \le 1$, and $x = \pi_{s,t}(z_s) \in M_t$. Otherwise, consider the finitely many points (r,b,u) of N with $r \le t$ and $z_{r-} \in b$. At each of them, either $u \le \min\{1, c_b(\eta_{r-})\}$, and then both processes exchange the spins along b and the path moves along b, or $u > \max\{1, c_b(\eta_{r-})\}$, and then nothing happens. Since $\eta_r(z_r)$ and $\xi_r(z_r)$ can change only at such points, and $\eta_0 = \xi_0$, we obtain $\eta_r(z_r) = \xi_r(z_r)$ for all $r \le t$, and in particular $\eta_t(x) = \xi_t(x)$.

We apply the Mecke equation for the Poisson process N [LPe18, Ch. 4], conditionally on η_0. Adding a point at time s to N changes neither $I_b(s)$ nor $\pi_{s,t}$, and the points of N in $[0,s)$ and

in $(s,t]$ are independent. Hence

$$P(x \in M_t) \le E \sum_{(s,b,u)\in N,\ s\le t} \mathbf{1}_{I_b(s)}(u) \sum_{w\in b} \mathbf{1}_{\{\pi_{s,t}(w)=x\}}$$
$$= \sum_{b\in\mathbb{B}_G} \sum_{w\in b} \int_0^t E\big[|I_b(s)|\big]\, P_w(X_{t-s}=x)\,\mathrm{d}s \le \gamma(x).$$

Since ξ_t has law $\nu_\beta T_t$, Step 2 of the proof of Theorem 8.1 gives $P(\xi_t(x)=+1)=\beta(x)$. Therefore

$$\big|P(\eta_t(x)=+1)-\beta(x)\big| \le P(x\in M_t) \le \gamma(x) \qquad (t\ge 0,\ x\in\mathbb{Z}^3).$$

Step 3: a stationary limit. Let μ_t be the law of η_t, and choose $T_n\to\infty$ such that $\frac{1}{T_n}\int_0^{T_n}\mu_t\,\mathrm{d}t$ converges weakly to some μ. The exchange dynamics with rates (8.2) is a Feller process [Li85, Ch. I], so μ is invariant, and in particular stationary, exactly as in Step 3 of the proof of Theorem 8.1. Since $\{\eta_x=+1\}$ is open and closed, Step 2 gives $|\mu(\eta_x=+1)-\beta(x)|\le\gamma(x)$. Together with Step 1, this proves all assertions except the last one.

Step 4: μ is not reversible. Suppose that μ is reversible, and let b be an edge of G. Reversibility says that $\int (h\circ\theta_b)\,c_b\,\mathrm{d}\mu=\int h\,c_b\,\mathrm{d}\mu$ for bounded measurable h. For bounded measurable f we apply this to $h:=f/(c_b\circ\theta_b)$. Since $\theta_b\circ\theta_b$ is the identity, (8.3) gives

$$\int f\circ\theta_b\,\mathrm{d}\mu=\int f\,e^{\Phi-\Phi\circ\theta_b}\,\mathrm{d}\mu. \tag{8.5}$$

Let $\Theta=\theta_{b_1}\circ\Theta'$ with $\Theta'=\theta_{b_2}\circ\cdots\circ\theta_{b_k}$ for edges $b_1,\dots,b_k$ of G, and suppose that $\int f\circ\Theta'\,\mathrm{d}\mu=\int f\,e^{\Phi-\Phi\circ\Theta'^{-1}}\,\mathrm{d}\mu$ for all bounded measurable f. Then $\int f\circ\Theta\,\mathrm{d}\mu=\int (f\circ\theta_{b_1})\,e^{\Phi-\Phi\circ\Theta'^{-1}}\,\mathrm{d}\mu$, and (8.5), applied to the function $f\,e^{\Phi\circ\theta_{b_1}-\Phi\circ\Theta'^{-1}\circ\theta_{b_1}}$, turns this into $\int f\,e^{\Phi-\Phi\circ\Theta^{-1}}\,\mathrm{d}\mu$, since $\Theta^{-1}=\Theta'^{-1}\circ\theta_{b_1}$. By induction on k we obtain, for every composition Θ of maps θ_b with $b\in\mathbb{B}_G$,

$$\int f\circ\Theta\,\mathrm{d}\mu=\int f\,e^{\Phi-\Phi\circ\Theta^{-1}}\,\mathrm{d}\mu. \tag{8.6}$$

Now let $v_+\in H_+$ and $v_-\in H_-$, and let $v_+=z_0,z_1,\dots,z_k=v_-$ be a path in G with distinct vertices. Let Θ be the composition of the exchanges along the edges

$$\{z_0,z_1\},\ \{z_1,z_2\},\ \dots,\ \{z_{k-1},z_k\},\ \{z_{k-2},z_{k-1}\},\ \dots,\ \{z_0,z_1\}.$$

Then Θ exchanges the spins at v_+ and v_- and leaves all other spins unchanged; in particular $\Theta^{-1}=\Theta$. Only the sets F that contain exactly one of v_+ and v_- contribute to $\Phi-\Phi\circ\Theta$, so $|\Phi-\Phi\circ\Theta|\le 2\big(j(v_+)+j(v_-)\big)$. Applying (8.6) to $f:=\mathbf{1}_{\{\eta_{v_-}=+1\}}$, for which $f\circ\Theta=\mathbf{1}_{\{\eta_{v_+}=+1\}}$, we obtain

$$\big|\mu(\eta_{v_+}=+1)-\mu(\eta_{v_-}=+1)\big|\le e^{2(j(v_+)+j(v_-))}-1.$$

Let $|v_\pm|\to\infty$ in $H_\pm$. The right-hand side tends to 0 by Step 1, whereas the left-hand side tends to $\rho_+-\rho_->0$ by Steps 1 and 3. Hence μ is not reversible. □

Remark 8.4. (a) Lammers and Toninelli [LT24] consider the heat-bath dynamics of an Ising model on a regular tree at large inverse temperature, with couplings that grow along the layers of the tree. They couple it with a zero-temperature majority dynamics that has a non-reversible stationary measure, and show that the non-reversibility survives. The proof above follows the same scheme in the opposite regime: the reference process is the exclusion process, i.e. the interaction is switched off at infinity instead of being made infinitely strong. The witness of non-reversibility in Step 4, the different limits of the density in the two ends, sits at infinity, where the coupling is good. Therefore no smallness of J is needed in any bounded region.

(b) Condition (8.1) is used to make Φ bounded, γ finite with $\gamma(x) \to 0$, and $j(z) \to 0$. We have not tried to optimise it.

(c) We do not know whether μ has finite energy. The argument of Step 5 of the proof of Theorem 8.1 does not apply, because η_t is not a random permutation of η_0.

(d) If J does not decay at infinity, for instance if it is a translation-invariant nearest-neighbour potential in each half-space, then ε_b does not tend to 0 and the bound of Step 2 is void. We expect non-reversible stationary measures with different chemical potentials in the two ends to exist also in this case, but the argument above does not show this. For interacting systems the density is then no longer the right witness: in a regime of phase coexistence, there are reversible measures whose densities in H_+ and in H_- differ.

(e) We do not see an analogue of Theorem 8.3 for spin flips. The construction rests on conservation: the densities in the two ends are maintained by the dynamics far away, and their difference drives a current through the bridge. For spin flips, the natural analogue, different phases in the two ends, is realised by Gibbs states, which are reversible. In [LT24] the non-reversibility comes instead from a flow along the orientation of a directed tree, which the authors relate to its non-amenability.

9. Remarks

Remark 9.1 (Relation to the proof of Holley and Stroock). For boxes, Proposition 4.2 is the inequality [HS77, (1.17), (1.18)]. Holley and Stroock obtain it from the time derivative of the free energy

$$F_\Lambda(\nu) := \sum_\sigma \nu(\eta_\Lambda = \sigma)\Big(\log \nu(\eta_\Lambda = \sigma) + \sum_{F \subseteq \Lambda} J_F \chi_F(\sigma)\Big),$$

computed with the formula of [MOP77]. The two derivations are related as follows. Let μ be any probability measure with $\mu(\eta_\Lambda = \sigma) > 0$ for all σ, and let $\mu_t = T_t^* \mu$. Then $\frac{\mathrm{d}}{\mathrm{d}t}\mu_t(\eta_\Lambda = \sigma)|_{t=0}$ is the left-hand side minus the right-hand side of (3.1). Using this in place of stationarity in the proof of Lemma 4.1 gives

$$\begin{aligned} 2\,\frac{\mathrm{d}}{\mathrm{d}t} F_\Lambda(\mu_t)\Big|_{t=0} = & -\sum_{k \in \Lambda} \Big[H_\Lambda\big(c_k\mu \,\big|\, (c_k\mu)\circ\theta_k^{-1}\big) + H_\Lambda\big((c_k\mu)\circ\theta_k^{-1} \,\big|\, c_k\mu\big)\Big] \\ & + \sum_{k \in \partial\Lambda} \sum_\sigma \Big[(c_k\mu)(\eta_\Lambda = \sigma) - (c_k\mu)(\eta_\Lambda = \sigma^k)\Big]\, r_k(\sigma), \end{aligned}$$

where $r_k(\sigma)$ is the bracket in the proof of Proposition 4.2. This is [HS77, (1.11)]. For stationary μ the left-hand side vanishes trivially. This does not force the entropy production to vanish: it only says that the entropy production equals the boundary term. The present proof works with the finite-volume dynamics instead and needs neither the free energy nor [MOP77].

The monotonicity (H2) and Pinsker's inequality (H3) play the roles of [HS77, (1.20)] and [HS77, (1.21)]. The essential difference lies in the way the estimates for different volumes are combined. In [HS77], the bounds for consecutive boxes lead to the recursion [HS77, (1.22)], which is solved by [HS77, Lemma 1.23]. Here the bounds for all level sets of a cut-off function are integrated at once (Theorem 5.1), and the dimension enters only through the capacity.

Remark 9.2 (Higher dimensions). Theorem 5.1 holds in every dimension. For $d \geq 3$ the network is transient [LP16, Ch. 2], and $\mathrm{cap}(B_n)$ is of order n^{d-2}. The upper bound $\mathrm{cap}(B_n) \leq Cn^{d-2}$ follows from $\psi(k) = \min\{1, (n/|k|)^{d-2}\}$ after the usual truncation to a finitely supported function. Hence, for every stationary μ, the left-hand side of (4.2) is at most Cn^{d-2}. A single box gives only $O(n^{d-1})$ (Remark 4.4).

The estimates of [HS77] give the same order. Let s_N be the left-hand side of [HS77, (1.22)], the sum over $n \le N$ of the entropy productions in the annuli $\Lambda_n \setminus \Lambda_{n-1}$ of [HS77]. Then [HS77, (1.22)] reads $s_N^2 \le c^2 N^{d-1}(s_N - s_{N-1})$ for a constant c. Whenever $s_{N-1} > 0$, this implies $1/s_{N-1} - 1/s_N \ge c^{-2}N^{1-d}$, and hence $s_N \le c^2\big(\sum_{n>N} n^{1-d}\big)^{-1} = O(N^{d-2})$. Now choose $N' \in [N, 2N]$ with $s_{N'} - s_{N'-1} \le s_{2N}/N$. The inequality preceding [HS77, (1.22)], applied in $\Lambda_{N'}$, together with [HS77, (1.20)], then bounds the entropy production in Λ_N by $O(N^{d-2})$. Neither argument decides whether non-Gibbsian stationary measures exist for $d \ge 3$. On regular trees, Lammers and Toninelli [LT24] constructed non-reversible stationary measures for stochastic Ising models whose couplings grow without bound.

Remark 9.3 (Graphs). Sections 3–5 use no property of $\mathbb{Z}^d$ beyond local finiteness. Let G be a connected, locally finite graph with vertex set V. Replace "$0 < |l-k| < L$" by "l is a neighbour of k in G", and define cap with unit conductances on the edges of G. Assume:

- $J_F \ne 0$ only if F is a clique of G, and $\sup_k \sum_{F \ni k} |J_F| < \infty$;
- each b_k depends only on the spins at the neighbours of k and takes values in $[b_-, b_+] \subset (0, \infty)$.

Replace the boxes B_n in Lemma 3.3 by an increasing sequence of finite sets exhausting V. The proofs above then show that, whenever simple random walk on G is recurrent, every stationary measure is a Gibbs state. No isoperimetric property of G is needed. By contrast, the recursion of [HS77], carried out on a graph, requires an exhaustion by finite sets Λ_n with $\sum_n |\partial\Lambda_n|^{-1} = \infty$. This is a condition of Nash-Williams type [NW59], which implies recurrence but is not necessary for it in general networks [LP16, Ch. 2].

Remark 9.4 (Positivity of the rates). The assumption $b_- > 0$ is used twice: for the positivity of the marginals (Lemma 3.2), without which the finite-volume dynamics is not defined, and for $K < \infty$. The example at the end of [HS77, §2] shows that the theorem fails without it.

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